

Assignment

Computer Networks LAB-01

Experiment

Study of LAN Topology, Network Device, and Address Configuration.

Objective:-

- 1) To study the LAN topology used in College campus.
- 2) To identify networking device & understand their functions.
- 3) To analyze wired & wireless Communication method used in the network.
- 4) To measure speed of network for wired & wireless connections.
- 5) To obtain MAC, IP, Address & subnet mask

Problem statement

Draw & identify the LAN topology of your organization.

Clearly label all components such as Router, switch, Access point, and end device (PCs, laptops, etc),

Soln:.

Identify LAN topology of college campus, clearly label all components such as router, switches & etc.

The college campus network observed this study consists of:

- * Layer-3 core switch in the server room.
- * Managed by POET switches.
- * Wireless Access point (APs)
- * Ethernet port in classroom & laboratories.

Network Topology:-

- ① Core layer
- ② Distribution layer
- ③ Access layer

These all are the network hierarchical layer.

Hybrid Hierarchical Topology which combines.

- ① Tree topology
- ② Extended star topology.
- ③ Ring Topology (server room)

Core Layer:

Core layer contains server Room and server room have firewall, Router, 2-L3+ switches patch Rack, server, Inverter etc.

Distribution layer

Distribution layer contains server Room and

The 4 switches are -

- (I) Main Build $\rightarrow$ R₇ (Room no-213, 2nd floor)
- (II) Main Build $\rightarrow$ R₁ (Room no-301, 3rd floor)
- (III) Main Build $\rightarrow$ R₆ (Room no-113, 1st floor)
- (IV) Main Build $\rightarrow$ R₂ (Room no-201, 2nd floor)

• This 4-switches contains $\rightarrow$

- (I) 18 port Gigabit easy smart switch with 16-port TL - S61218-PI

- (II) Sc 34 28 XMP $\rightarrow$ 24 Port Gigabit & 4 port 10GE-SFP, L2+ Managed switches with 24 port POE +

$\rightarrow$ Now let discuss these 2 above components

Now TP Links TL - SR1218MPE

• Linkely used for: CCTV camera, wifi access point etc.

- 18 port Gigabit "Easy Smart" switch
- 16 port support PoE +

② TR-Link Omada S93428 XHP

- Higher end managed layer 2+ switch
- 24 PoE port, uplinks, 10G SFP
- usually acts as a main distribution for each floor

Access Layer :-

Access layer contain 3 switches i.e

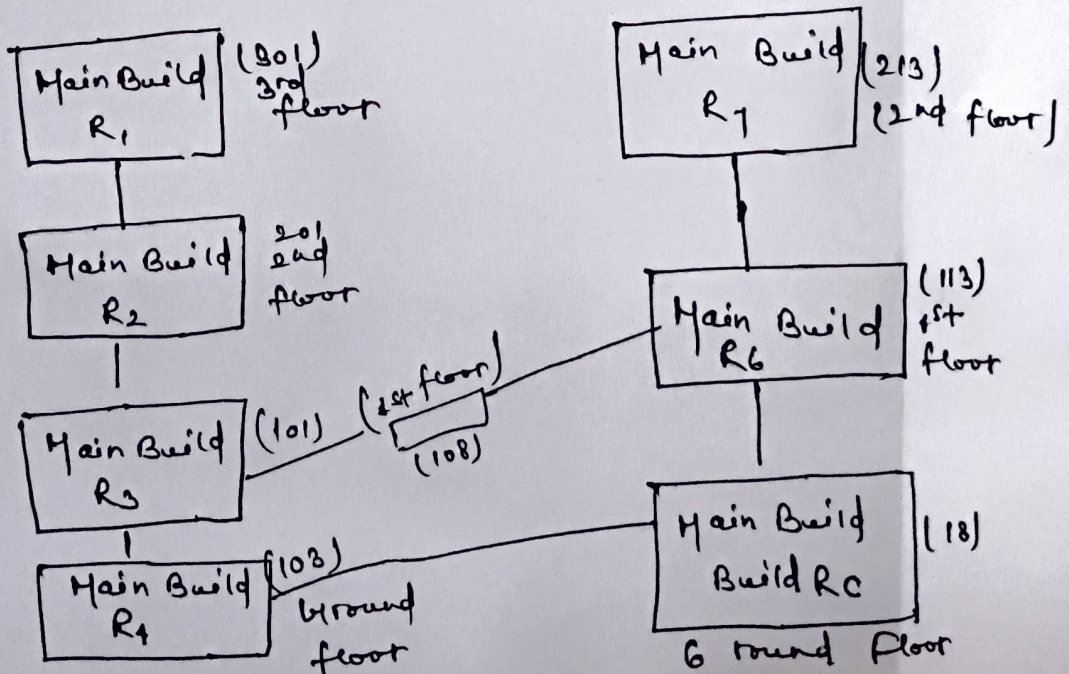
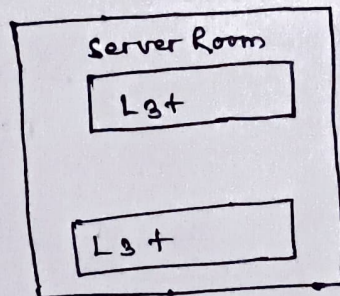
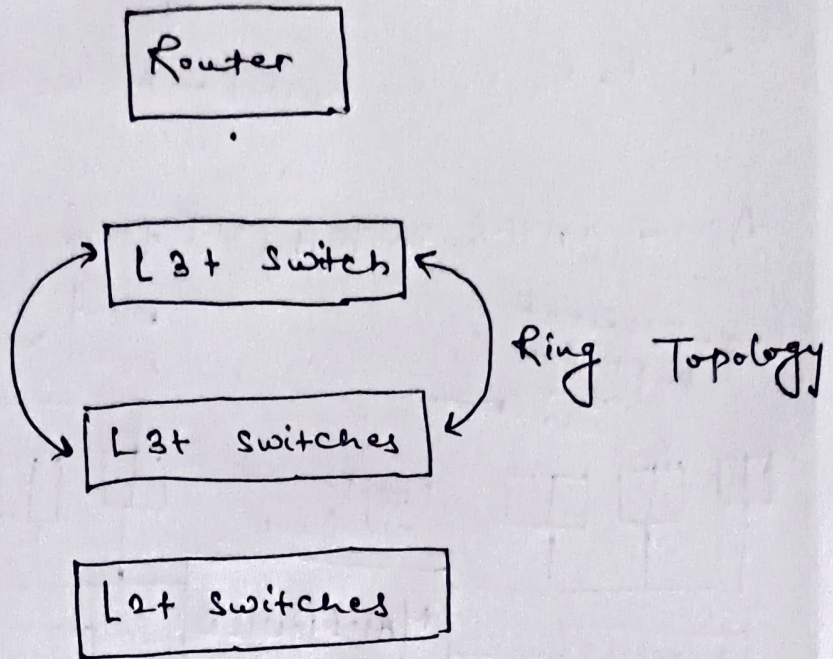
- (i) Main Build R₁ - [Room no - 301, 3rd floor]
- (ii) Main Build R₄ - [Room no. Ground floor]
- (iii) Main Build R₅ - [Room no - 18, Ground floor]

- Every switches act as access layer which helps to provide network connectivity to all classroom, lab, faculty Room etc.

• Access layer directly connect :-

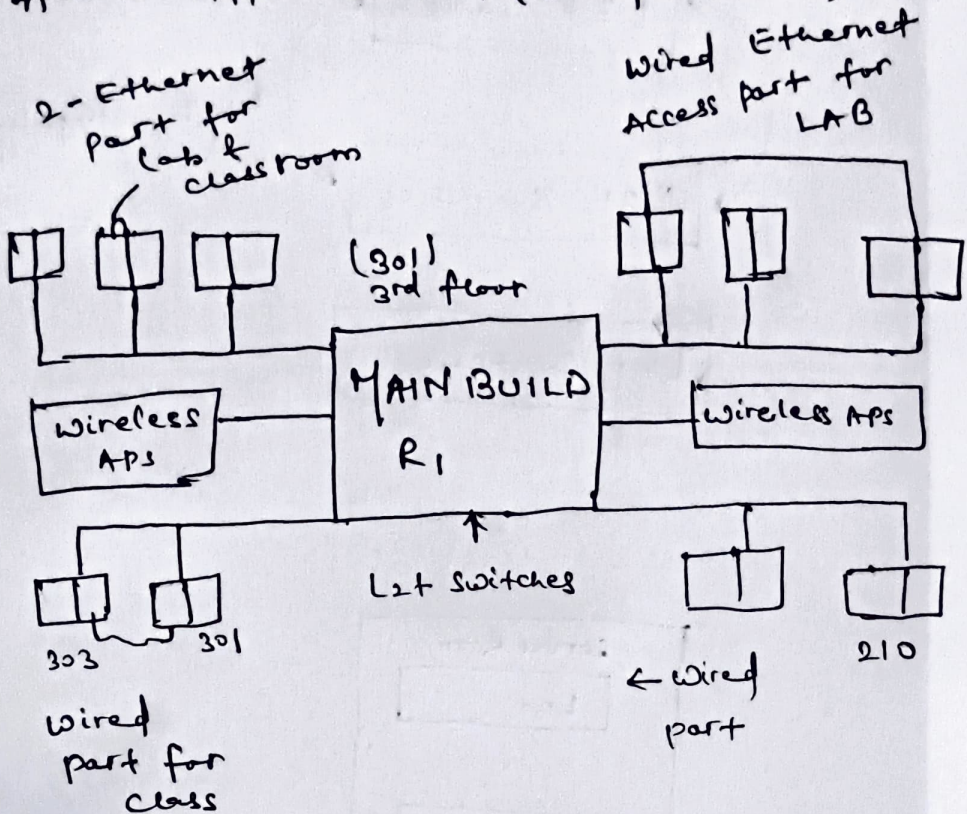
- (i) Wireless Ap
- (ii) Ethernet port
- (iii) Lab system
- (iv) Outdoor wireless Ap (ex - 018)

→ server Room Architecture / Topology :-



- R_1, R_7, R_2, R_6 → Distribution layer
- Server Room → Core layer
- $R_1, R_3, R_4, R_5, R_6, R_7$ → Access layer.

Access LAYER SWITCH (3rd floor - 301)



Windows PowerShell

Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! <https://aka.ms/PSWindows>

PS C:\Users\Divya> ipconfig /all

Windows IP Configuration

Host Name : DESKTOP-A2N2J7F
Primary Dns Suffix :
Node Type : Hybrid
IP Routing Enabled. : No
WINS Proxy Enabled. : No

Ethernet adapter Ethernet:

Media State : Media disconnected
Connection-specific DNS Suffix :
Description : Intel(R) Ethernet Connection (6) I219-LM
Physical Address. : E8-D8-D1-F4-E6-E5
DHCP Enabled. : Yes
Autoconfiguration Enabled : Yes

Wireless LAN adapter Local Area Connection* 1:

Media State : Media disconnected
Connection-specific DNS Suffix :
Description : Microsoft Wi-Fi Direct Virtual Adapter
Physical Address. : 5C-80-B6-F2-F8-BA
DHCP Enabled. : Yes
Autoconfiguration Enabled : Yes

Wireless LAN adapter Local Area Connection* 2:

Media State : Media disconnected

Physical Address : 5C-80-B6-F2-F8-BA
DHCP Enabled : Yes
Autoconfiguration Enabled : Yes

Wireless LAN adapter Local Area Connection* 2:

Media State : Media disconnected
Connection-specific DNS Suffix :
Description : Microsoft Wi-Fi Direct Virtual Adapter #2
Physical Address : 5E-80-B6-F2-F8-B9
DHCP Enabled : No
Autoconfiguration Enabled : Yes

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix :
Description : Intel(R) Wi-Fi 6 AX200 160MHz
Physical Address : 5C-80-B6-F2-F8-B9
DHCP Enabled : Yes
Autoconfiguration Enabled : Yes
Link-local IPv6 Address : fe80::953e:cb74:2415:1f35%12(Preferred)
IPv4 Address : 10.85.20.253(Preferred)
Subnet Mask : 255.255.248.0
Lease Obtained : 21 July 2026 10:28:02
Lease Expires : 22 July 2026 08:45:55
Default Gateway : 10.85.16.1
DHCP Server : 10.85.16.1
DHCPv6 IAID : 123502774
DHCPv6 Client DUID : 00-01-00-01-30-6B-6B-0E-E8-D8-D1-F4-E6-E5
DNS Servers : 8.8.8.8
 : 1.1.1.1
 : Enabled
NetBIOS over Tcpip : Enabled

Sublime



Processes



Performance



App history



Startup apps



Users



Details



Services

Performance

Run new task ...



CPU

23% 1.52 GHz



Memory

5.9/7.8 GB (76%)



Disk 0 (C: D:)

SSD (SATA)
72%



Wi-Fi

Wi-Fi

S: 0 R: 72.0 Kbps



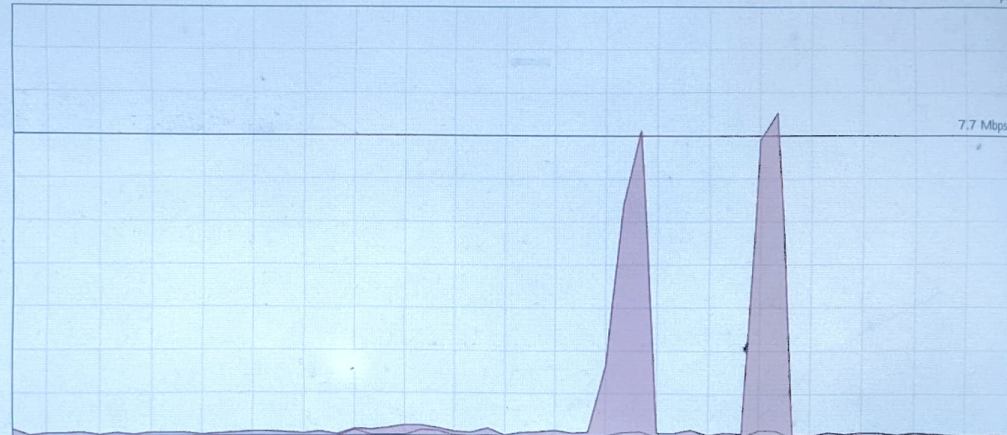
GPU 0

Intel(R) UHD Graphics ...
1%

Wi-Fi

Throughput

Intel(R) Wi-Fi 6 AX200 160MHz



60 seconds

Send

0 Kbps

Receive

72.0 Kbps

Adapter name:

Wi-Fi

SSID:

GECK-Campus-WiFi

Connection type:

802.11ac

IPv4 address:

10.85.20.253

IPv6 address:

fe80::953e:cb74:2415:1f35%12

Signal strength:



Settings

Windows PowerShell

Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! <https://aka.ms/PSWin>

PS C:\Users\Divya> ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

Media State : Media disconnected
Connection-specific DNS Suffix . :

Ethernet adapter Ethernet 2:

Connection-specific DNS Suffix . :
Link-local IPv6 Address : fe80::efef:7ea:34ec:1b15%42
IPv4 Address. : 10.94.48.1
Subnet Mask : 255.255.255.0
Default Gateway : 10.94.48.85

Wireless LAN adapter Local Area Connection* 1:

Media State : Media disconnected
Connection-specific DNS Suffix . :

Wireless LAN adapter Local Area Connection* 2:

Media State : Media disconnected
Connection-specific DNS Suffix . :

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . :
Link-local IPv6 Address : fe80::953e:cb74:2415:1f35%12
IPv4 Address. : 10.85.20.253
Subnet Mask : 255.255.248.0
Default Gateway : 10.85.16.1

Ethernet adapter Bluetooth Network Connection:

Media State : Media disconnected
Connection-specific DNS Suffix . :

PS C:\Users\Divya> |

→ Difference between wired & wireless speed?

Wired Network	Wireless Network
• Faster communication • stable connection • low latency • Dedicated physical medium	• Slower communication • Affected by distance & interference • Higher latency • Shared radio medium

→ Ethernet provides direct physical communication using network cables, resulting in higher speed & stable performance.

→ Wireless connections used radio communication waves which are affected by interference, distance, obstacles & multiple connected users.

Q. Role of Subnet Mask in network?

A subnet mask identifies the network portion & host portion of an IP address. It helps devices determine whether another device belongs to the same network or different network.

• It is important for:

- (I) Network segment
- (II) efficient routing
- (III) Communication between devices.

OBSERVATION

During the study of the college network infrastructure:-

- (I) A centralized server room was used of network backbone.
- (II) Two L3 switches were connected in redundant cycle's structure.